Problem Statement-1: Edge AI for Solving Day-to-Day Air Pollution Problems with Arduino UNO Q

Description: Air quality monitoring today is mostly cloud dependent. Sensor nodes push raw readings to a server, and every decision waits on connectivity, bandwidth and a subscription. Citizens are left with a single AQI number for an entire city, which says nothing about the road they walk on, the classroom they sit in, or the kitchen they cook in. The gap is not sensing. It is interpretation at the place where the pollution is actually happening. Edge AI closes that gap by running the model on the device itself, so the node can classify, forecast and advise even when the network is down.

> **Mandatory Hardware Requirement:** The solution must be built on the **Arduino UNO Q**. All AI inference must run on the board itself. Cloud APIs and external AI services may not be used for the core decision-making of the project. Sensors, actuators, displays and communication modules are open to the team's choice, but the UNO Q must be the computing platform of the submitted prototype.

Challenge:

Using the Arduino UNO Q as the compute platform, design an Edge AI node that:

- Senses PM2.5, PM10, CO, CO_2, NO_2, VOCs, temperature and humidity, and runs all inference locally with no cloud round trip.
- Forecasts the next few hours of hyperlocal air quality from the node's own sensor history using a small on-device time series model.
- Identifies the likely pollution source from sensor signatures, for example traffic, garbage burning, construction dust, cooking smoke or crop residue burning, using sensor fusion and optionally a camera or microphone input.
- Converts readings into an action the user can take that day, such as changing a walking route, timing the opening of windows, running an exhaust fan, or moving an outdoor activity indoors.
- Keeps working through power cuts and network outages, storing data locally and syncing only compressed summaries when a link becomes available.
- Scales to multiple nodes across a campus, colony or ward so that a street-level pollution map can be built without any per-node cloud cost.

Expected Outcome:

- Hyperlocal air quality intelligence that works offline and costs nothing to run per node.
- Actionable guidance for daily life instead of a raw number the user cannot interpret.
- Source attribution that helps a resident welfare association, school or municipal body fix the cause and not just record the effect.
- Privacy preserving operation, since raw audio, images and location never leave the device.
- A deployable low-cost reference design that schools and local bodies can replicate.

Technical Feasibility:

- Board: Arduino UNO Q, which pairs a Qualcomm Dragonwing QRB2210 quad-core Cortex-A53 processor running Debian Linux with an STM32U585 microcontroller for real-time sensing, giving both AI capability and deterministic sensor timing on one board.
- Sensors: PMS5003 or SPS30 for particulates, MQ135 and MiCS-6814 for gases, SCD41 for CO_2, BME688 or DHT22 for temperature, humidity and VOC index. Modulino nodes over I2C for rapid prototyping.
- Software: Arduino App Lab with its Python and C++ bricks, allowing the Linux side to run the model while the MCU side handles acquisition and actuation.
- Models: quantised TensorFlow Lite or ONNX Runtime models, gradient boosting for classification, a compact LSTM or temporal convolutional network for forecasting, and optionally a small language model running fully on device for the advisory layer.
- Connectivity: on-board Wi-Fi and Bluetooth, with LoRa or GSM as an add-on for outdoor nodes.
- Power: USB-C, or a solar panel with a LiFePO4 pack for a fully off-grid outdoor node.